CSIRO PUBLISHING

Marine and Freshwater Research, 2010, **61**, 642–650 www.publish.csiro.au/journals/mfr

Fish and sessile assemblages associated with wind-turbine constructions in the Baltic Sea

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Abstract. Offshore wind farms are being built at a high rate around the world to meet the demand for renewable energy. We studied fish and sessile communities on and around offshore wind-turbine foundations in the southern Baltic Sea, 7 years after construction, using visual census techniques to determine how fish, sessile-invertebrate and algal communities are affected by the introduction of such structures. Fish assemblages were dominated by two-spotted gobies (*Gobiusculus flavescens*) that were found in large shoals in close association with the vertical surface. At the seabed, close to the foundation, the black goby (*Gobius niger*) was recorded in large numbers. The most obvious difference in fish densities was found between wind-power foundations extending through the entire water column and the surrounding open waters. Fouling assemblages on the vertical foundation surfaces and at the seabed just below differed from those at the seabed further away by having higher coverage of blue mussel (*Mytilus edulis*) and less algal growth. The results from the present study suggest that the introduction of offshore wind turbines in marine waters could have a positive effect on fish numbers and the presence of sessile invertebrates.

Additional keywords: artificial reef, disturbance, habitat structure, reef effect, wind power.

Introduction

The development of offshore wind power has increased the occurrence of artificial structures in the marine environment. This may create new habitat opportunities for fish and fouling organisms. However, little scientific information exists on how marine organisms are affected by such structures (but see Wilhelmsson *et al.* 2006*a*; Zettler and Pollehne 2006; Maar *et al.* 2009). A wind-turbine foundation is commonly vertical and extends through the whole water column, providing a hard substrate above the seabed in an area that previously was open water. By adding hard substrates, a reef effect may be achieved. To examine the effect of offshore wind-power development, comparisons could be made with oil-rigs (Helvey 2002), bridge-pilings (Qvarfordt *et al.* 2006) and jetties (Rilov and Benayahu 2000). However, turbines may have different surface materials, structure and locations, and are commonly built in wind farms that can contain several units in one area. Wind farms create an area with patches of hard substrates spread over a larger area, which potentially provides marine organisms with an extended home range (Kramer and Chapman 1999). In addition, wind farms are usually located in shallow waters (<20 m) in areas that previously had received little anthropogenic disturbance. Wind turbines can also influence marine life by the sound emitted during construction and operation (Wahlberg and Westerberg 2005; Andersson *et al.* 2007) as well as the electromagnetic fields generated by the cables connecting the turbines with the power grid on land (Gill and Kimber 2005; Öhman *et al.* 2007). Thus, for a better understanding of offshore wind-power effects on marine life, investigations are needed *in situ*.

The structural properties of wind-turbine foundations may provide unique habitat opportunities that could benefit a range of species. Because the foundations penetrate the whole water column, their presence could increase the likelihood that fish and invertebrate larvae will encounter a suitable habitat for settlement (Neira 2005). Further, the foundation will change water movement, with a reduced velocity at the leeward side (Guichard *et al.* 2001). This change in the current regime with water that has not been filtered by other organisms may promote food availability for both filter-feeding invertebrates on the foundation (Fréchette *et al.* 1989; Perkol-Finkel and Benayahu 2007) and plankton-feeding fish (Rilov and Benayahu 1998; Neira 2005). Likewise, migratory fish could be attracted by the structures as they may find food and protection.

In the only published study examining reef effects of offshore wind farms on local fish assemblages, high densities of benthic and semi-pelagic fish were found to associate with the foundation, and at the seabed close to the foundation (Wilhelmsson *et al.* 2006*a*). Settlement opportunities for larvae, and available food and shelter, were suggested to be important factors. Studies of the effectiveness of artificial reefs in terms of increasing fish numbers have been inconclusive (Powers *et al.* 2003; Brickhill *et al.* 2005). The regulating mechanisms, as suggested by Bohnsack (1989), could be either an attraction affecting the behavioural response of existing fish (i.e. redistribution) or an increase in biomass (new production) as a result of settlement of fish larvae in response to a new habitat.

The aim of the present study was to examine how the introduction of offshore wind-power foundations would

 10.1071/MF09117 1323-1650/10/060642

influence fish and sessile invertebrate communities 7 years after construction. We predicted that the wind-turbine foundations would provide new habitat opportunities for fish and invertebrates. This would have a local positive effect on fish numbers, both juveniles and adults, compared with the surrounding seabed and open waters. Communities of sessile organisms would increase because vertical substratum for settlement would be made available; however, these were hypothesised to differ from those on the natural seabed. Comparisons were made with an early study in the area, to give an indication of changes over time (cf. Wilhelmsson *et al.* 2006*a*; Wilhelmsson and Malm 2008).

Materials and methods

Study area

The study was carried out in Utgrunden wind farm (N56 22.416, E16 15.401) which contains seven wind turbines that were constructed in 2000. The wind farm is located in the Kalmar Strait (southern Baltic Sea), with similar brackish water conditions throughout the strait. The wind turbines stand on a submerged glacial boulder ridge, 12 km from the Swedish mainland and 8 km from the island of Öland (Fig. 1). The wind-turbine foundations are steel monopiles, with a diameter of 3 m. They are driven into the ridge and stand ~350 m apart, at a depth of 7–9 m. The steel surface is painted with a glass-flake reinforced polyester coating, creating a smooth surface. The seabed around the foundations consists of patches of small boulders, sand and gravel. The four control sites (A–D, Fig. 1) were located at the extension of the same ridge, at a distance of 1.0, 2.0, 7.8 and 9.5 km, respectively, north of the wind farm (species assemblages among the four control sites were statistically similar – see Results). The choice of control sites was determined by the availability of similar substratum characteristics in the area. To compare with a structure that resembles a wind-turbine foundation, a lighthouse (Utblicken) was sampled, 2.6 km north of the wind farm. The lighthouse is 50 years old and has a concrete foundation at 9 m depth.

Water temperature in the strait varies between 0 and 20°C, depending on the season. During cold winters, the strait may be covered by ice. Salinity of the surface water varies between 6.5 and 7.5. The direction of the current in the strait fluctuates between north and south, with an average speed of 0.1 m s^{-1} in the summer and 0.3 m s^{-1} during autumn and winter (Svensson and Ambjörn 2002). During the present survey, water temperature, wave height and wind speed varied between 15 and 18°C, 0 and 1.5 m and 3 and 14 m s^{-1}, respectively. The wind direction fluctuated between north and south-west. Underwater visibility, visually estimated by divers, was 2–6 m.

Field method

The survey was carried out on six of seven turbines (one wind turbine was not in use during the study period and was therefore not surveyed) from 22 to 30 August 2007. Fish and sessile communities were examined by scuba diving within four 10-m strip transects at each site, delineated by a measuring tape. Fish

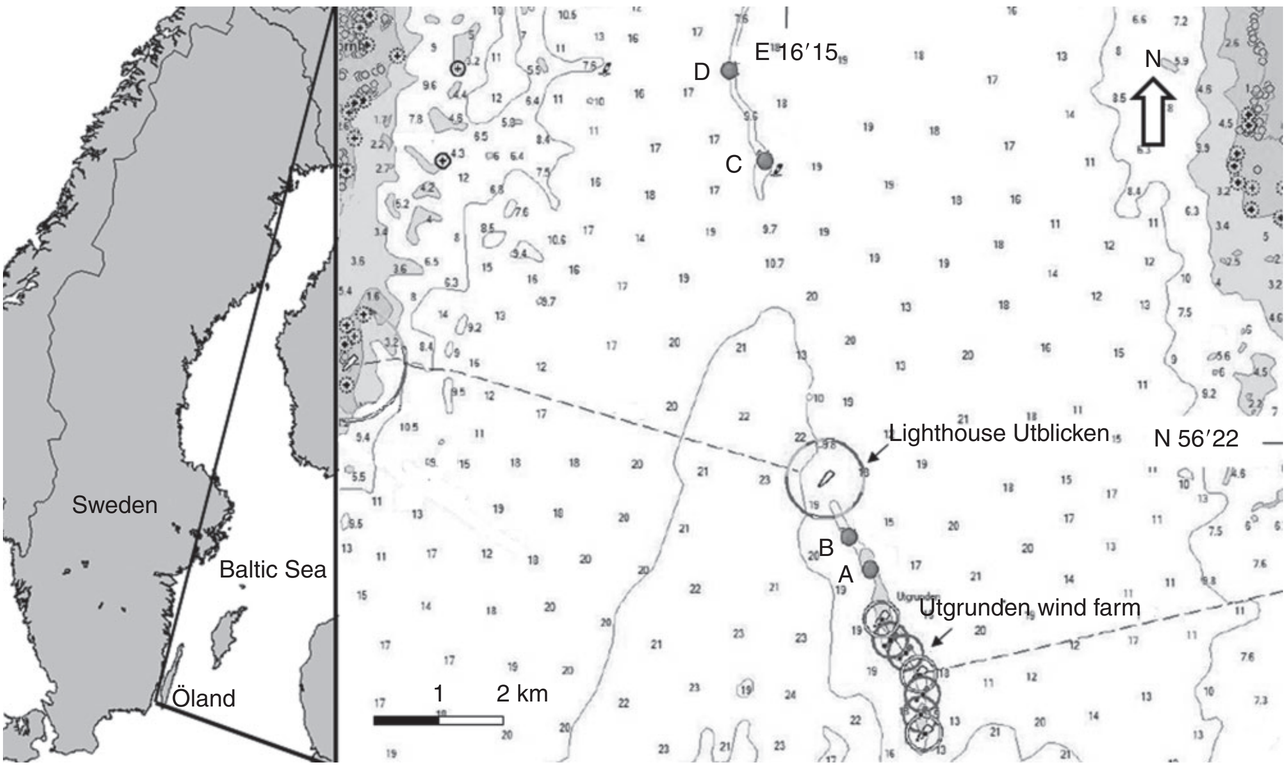

Fig. 1. Map of Sweden and the Kalmar Strait, showing the Utgrunden wind farm, the lighthouse Utblicken and control sites (A–D). Depth numbers are in metres.

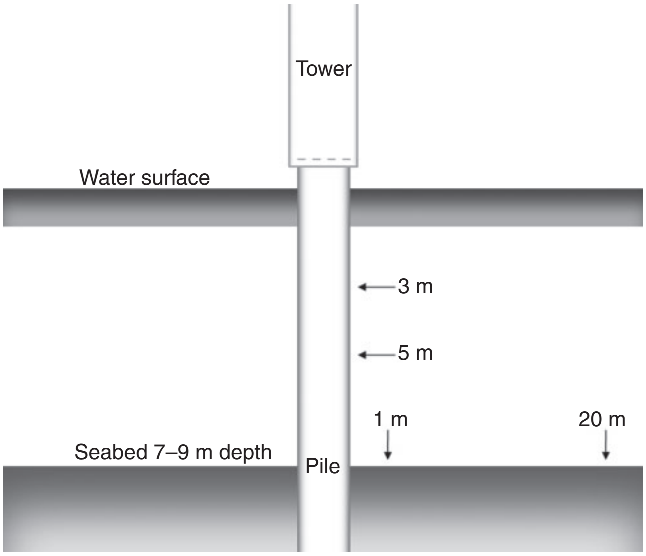

Fig. 2. Layout of strip transects at the wind turbine where 1 and 20 m denote the distance from the foundation perimeter where the linear transects were taken, and 3 and 5 m give the depth of transects encircling the foundation. Drawing is not to scale.

were counted within a belt of 10×1 m, up to 1 m above the substratum. On the vertical surface of the foundations, the areas were denoted by laying the tape horizontally at the depths of 3 and 5 m, following the vertical structure. At the seabed, the strip transects were taken at the northern side of the foundations, with the centre of the tapes fixed at 1 and 20 m from the foundation wall (Fig. 2). Fish counts were conducted 5 min after the transect tape was put in place.

Substratum characteristics and percentage coverage of habitat-forming organism within transects were estimated and grouped into mussels, red algae, hydroids, barnacles and free space (non-colonised areas on the seabed and on the vertical surfaces). The same transect technique and general survey was applied at four control sites and at the lighthouse; at the control sites, transects were laid as if at an imaginary wind-turbine foundation.

A general visual survey on and around the foundations of each wind turbine, within a 20-m radius from the turbines, was also performed to obtain qualitative data on the fauna and flora. In addition, scrapes of samples from both the seabed and foundation were taken and analysed in the laboratory when identification of the species could not be conducted in the field.

Because blue mussels in the Baltic Sea are described as either *Mytilus edulis* or *M. trossulus*, or a hybrid of both (Śmietanka *et al.* 2004), we list them here as *Mytilus* spp. All fish were identified to the lowest taxonomic level possible. All *Pomatoschistus* species were grouped together as sand gobies (*Pomatoschistus minutus*), acknowledging that some individuals could be common gobies. Fish species were also divided into separate life stages (i.e. juveniles and adults), depending on the size. Species were characterised as juveniles according to the following criteria: two-spotted goby (*Gobiusculus flavescens*) <25 mm (Folkestad 2002), black goby (*Gobius niger*) <50 mm (Vesey and Langford 1985), viviparous eelpout (*Zoarces viviparus*) <170 mm (Kosior and Kuczyński 1997) and sand goby (*P. minutus*) <30 mm (Leitão *et al.* 2006).

Data analysis

We tested whether the wind-turbine foundations would function as new habitat for juvenile and adult fish. Dependent variables were total abundance (N), and abundance of juveniles (J) and adults (A). This concerned all species combined and the two most common species (i.e. two-spotted goby and black goby). Because two fish transects that were taken on the vertical surface of the foundations at 3- and 5-m depth were similar ($P > 0.05$), data were pooled and presented as mean values. Transects on vertical structures are hereafter named W for wind turbines, C for control sites and LH for the lighthouse.

To compare data from the wind-turbine foundation with those from the nearby seabed, the non-parametric Wilcoxon signed ranks test was applied to transects on the turbine (W) and on the seabed, at a distance of 1 m from the turbine (W1), and at a distance of 20 m from the turbine (W20). The two transects 1 and 20 m away from the imaginary foundation (C1 and C20) at the four control sites were compared by the same method.

Transects at control sites (C1–C20) did not differ significantly ($P > 0.2$) in the pair-wise comparison of densities of juveniles, adults and total density. As a consequence, the control transects C1 and C20 at each site were used to compare fish data with those on the wind-turbine transects in the following analysis. To examine whether the wind-turbine structure supported different species, life stages and community composition of fish from those in areas of similar seabed substratum and depth, transects at the four control sites were compared with turbine transects (W to C, W1 to C1 and W20 to C20). The control sites were treated as independent sites, acknowledging the difference in spatial scale compared with the turbine sites. Hence, the replication was six for wind turbines and four for controls in all analyses, and the data were compared with the non-parametric Kruskal–Wallis one-way analysis of variance because the homogeneity of variances was not achieved.

All univariate data were analysed by using SPSS13.0 statistical software (SPSS Inc., Chicago, IL, USA).

Fish assemblage composition was examined with PRIMER-E v6 (Plymouth, UK) ANOSIM one-way permutation test, based on rank similarity matrices (Clarke 1993). Data was $\log(x + 1)$-transformed to decrease the impact of dominant species and Bray–Curtis similarity measures were applied (Clarke 1993). To identify which species contributed the most to the observed similarity, SIMPER (similarity percentage procedure) was used.

The habitat variables (i.e. habitat-forming organisms) on the seabed and non-colonised substratum (bare sand, mud, gravel or rocks) and fouling assemblages on foundation surfaces were analysed with the multivariate ANOSIM on square-root-transformed data. First, control seabed transects (C1 and C20) were analysed for differences between sites, using the two transects as subsamples. Then turbine and control seabed transects (W1 and C1, W20 and C20) were compared for differences in community structure. The comparison W–C could not statistically be carried out due to lack of vertical substrate on control sites.

Results

Habitat

There was vertical zonation down the foundation from the water surface to the seabed and around the base. Common sessile organisms from the surface down to 3 m were filamentous green algae *Cladophora* sp., the red alga *Ceramium tenuicorne* and the barnacle *Balanus improvisus*. Further down (3–8 m), the blue mussel together with two species of red algae, *Polysiphonia fucoides* and *Rhodochorton purpureum*, dominated. The marine hydroid *Laomedea loveni* occurred in small numbers. There were similar fouling organisms at the two sampling depths (3 and 5 m) on the foundation (Fig. 3*a*); multivariate analysis revealed no significant difference between the assemblages (ANOSIM; $r = -0.096$, $P = 0.814$). The sessile fauna on the foundation differed from that on the seabed transects (ANOSIM; W–W1 $r = 0.748$, $P = 0.006$; W–W20 $r = 0.685$, $P = 0.004$), especially barnacle and hydroid coverage. Beneath the foundation, a dense coverage of blue mussels was found as well as small turfs of red algae (*P. fucoides*, *R. purpureum* and *Rhodomela confervoides*); the opposite pattern occurred at 20-m distance where red algae dominated the seabed (Fig. 3*b*). These differences in the composition of sessile communities were also evident in the analysis when comparing transects 1 m (W1) and 20 m (W20) away from the foundation (ANOSIM; $r = 0.367$, $P = 0.011$).

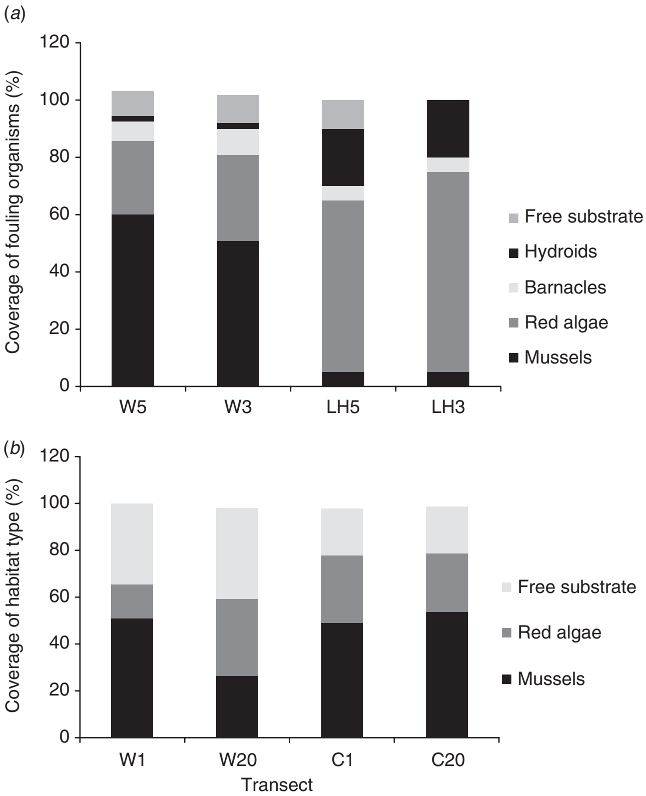

Fig. 3. Average coverage per 10 m^3 of habitat-forming sessile invertebrates and algae (*a*) on the vertical foundations and (*b*) at the seabed. Transects are: W/C = transect on the foundation/control at 3- and 5-m depth, W1/C1 and W20/C20 = transect at the bottom at a 1- and 20-m distance from the real or imaginary wind-turbine foundation, respectively.

The control sites contained similar habitat-forming organisms and non-colonised substrate as did seabed transects around the wind turbines (Fig. 4*b*). No difference in coverage between the four control sites and turbine transects was seen (ANOSIM; W1–C1 $r = 0.234$, $P = 0.100$; W20–C20 $r = 0.226$, $P = 0.076$). Notably, there was a greater difference in assemblage structure (greater *R*-value) between the two transects close to the wind-turbine foundation than between the transects close to the wind-turbine foundation and those in the distant control areas.

The fouling assemblage on the lighthouse concrete foundation was different from that on the wind-turbine foundations, with a dissimilar coverage ratio, especially for hydroids (Fig. 3*a*). No statistical comparison was made because of a shortage of replication of lighthouse surfaces.

Fish

Approximately 8000 fish specimens were recorded, including nine species, around the wind-turbine foundations. The most common species was the two-spotted goby (*Gobiusculus flavescens*) that constituted ~70% of the total fish abundance. They occurred in large (>1000 individuals) and small (<10 individuals) shoals, as well as single individuals. Other species recorded were the black goby (*Gobius niger*), sand goby (*Pomatoschistus minutus*), rock gunnel (*Pholis gunnellus*) and viviparous eelpout (*Zoarces viviparus*). Of these species, black gobies were found in highest numbers (Table 1). Juvenile fish comprised 14% of the total abundance, sharing distribution patterns similar to those of the adults. Gravid two-spotted gobies were observed on three wind turbines.

The majority of recorded fish, including both juveniles and adults, were found on and close to the foundation; however, there were large variations (Fig. 4*a*), resulting in non-significant differences in total fish density between transects around the wind turbines (Table 2). Differences occurred in numbers of two-spotted gobies and black gobies (total density, juveniles and adults) when transects on the turbines (W) were compared with W1 and W20; however, there were no differences between the W1 and W20 transects (Table 2, Fig. 4*b*).

Black gobies were more abundant closer to the foundation (W1) than further away (W20) (Fig. 4*c*), although the Wilcoxon's test did not show significant differences when comparing the means of these transects around the turbines (Table 2). No black gobies were found on the wind-turbine foundation, which explains the difference seen in the comparisons with seabed transects (Table 2). When comparing the overall average abundance of black gobies at the control-site transects (C1 and C2) and the seabed transects within the turbine area (W1 and W20), the abundances were similar (wind-turbine area: 19.1, s.e. 2.7; controls: 19.9, s.e. 1.3). Similar densities of black gobies were found around the lighthouse (23.5, s.e. 1.5); however, these data were based on only two transects.

The capacity of wind-turbine foundations to enhance fish numbers was only seen in the comparison between the water column of the control sites and that of the wind-turbine foundation (Table 2). However, the lower densities of juvenile fish at C20 than at W20 resulted in a non-significant difference ($P = 0.10$, Kruskal–Wallis). The structure of the fish assemblage differed

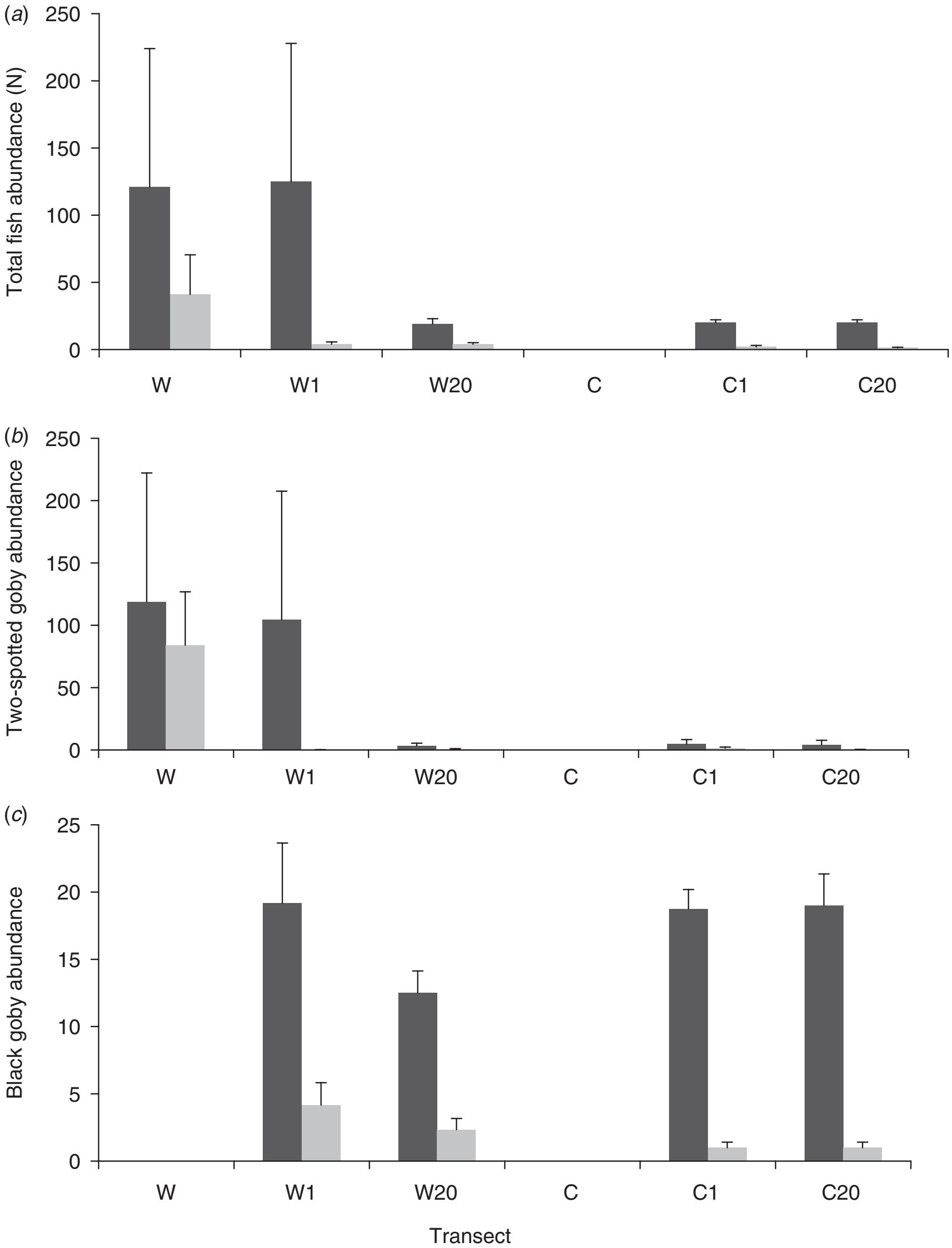

Fig. 4. Mean fish abundance (+s.e.) per $10\,m^3$ for (*a*) all species, (*b*) two-spotted goby and (*c*) black goby individually, at the turbine ($n = 6$) and control sites ($n = 4$), divided by life stage; adults (dark grey) and juveniles (light grey). Abbreviations are as in Fig. 3.

between the transects on the turbine and the seabed (ANOSIM; W–W1 $r = 0.743$, $P = 0.002$; W–W20 $r = 0.869$, $P = 0.002$), but not between the two seabed transects, W1 and W20, which showed similar fish assemblages ($r = -0.074$, $P = 0.61$).

When turbine and control transects were compared, fish assemblages differed between the transects on the turbine and the water column at the control sites (W–C $r = 1$, $P = 0.005$). There were no differences between the turbine seabed transects and the controls transects (W1–C1, W20–C20; $R < 0.150$, $P > 0.10$). The SIMPER analysis revealed that although a higher density of the two-spotted goby was found in the turbine area, the similar distribution of black gobies evened out that difference.

General visual observations of the areas during sampling revealed some interesting patterns. The two-spotted gobies were commonly found around the cables as well as the ladders attached to the foundations. Large shoals of two-spotted gobies were seen at all turbine sites, although not always at transect depths. Different size classes were usually separated in different shoals. Schools of juveniles were always accompanied by adults at the outskirts of the groups. Longspined bullheads (*Taurulus*

Table 1. Mean fish abundance and standard error (s.e.) per 10 m^3 of juveniles (*J*) and adults (*A*) at the wind turbines (W, W1, W20), control sites (C, C1, C20) and the lighthouse (LH, LH1, LH20)
Data are based on a sample size of six turbines, four control sites and one lighthouse

Common name	Scientific name	Wind turbine												Control										Lighthouse					
		W				W1				W20				C		C1				C20				LH		B1		B20	
		J	s.e.	*A*	s.e.	*J*	s.e.	*A*	s.e.	*J*	s.e.	*A*	s.e.	*J*	*A*	*J*	s.e.	*A*	s.e.	*J*	s.e.	*A*	s.e.	*J*	*A*	*J*	*A*	*J*	*A*
Two-spotted goby	*Gobiusculus flavescens*	48.8	31.7	121.3	112.0	0.2	0.2	104.5	103.1	0.7	0.4	3.3	2.1			1.0	1.0	0.3	0.3					180	1500	50	60		
Black goby	*Gobius niger*					4.2	1.7	19.2	4.5	2.3	0.9	12.5	1.6			1	0.4	18.8	0.7	1.0	0.2	19.0	2.3			2	20	3	22
Sand goby	*Pomatoschistus minutus*							1.3	1.3	0.7	0.5	2.8	2.3																
Viviparous eelpout	*Zoarces viviparus*																	1.0	1.0			1.0	1.0						
Rock gunnel	*Pholis gunnellus*			0.1	0.1																								
Total fish abundace		48.8	31.7	121.4	103.0	4.4	1.65	125.0	102.9	3.7	1.1	18.6	3.9	0	0	2.0	1.0	19.1	2.1	1.0	0.5	20.0	2.1	180	1500	52	80	3	22

bubalis) were seen on rocks or hiding at the base of the foundation and several male black gobies, in spawning colours, were observed close to each other underneath rocks and in crevasses around the turbines and at the controls. Flounders (*Platichthys flesus*) appeared on sandy areas, and broad-nosed pipefish (*Syngnathus typhle*) and three-spined sticklebacks (*Gasterosteus aculeatus*) were seen in the water column. Overall, more species were recorded around the wind turbines than at the control sites. The survey at the lighthouse Utblicken showed similar fish density and assemblage structure as in transects taken at the turbines (Table 1).

Discussion

The present study showed that offshore wind-power developments alter the nearby areas and create new habitat opportunities for fish, algae and invertebrates. Fish occurred in higher densities at the wind turbines than at control sites, and the fouling assemblages on the foundations were different from those on the seabed. Because the wind-power foundations penetrated the whole water column, comparisons should also be made with the surrounding open water. Indeed, the added habitat locally increased the presence of fish and sessile organisms that were otherwise not seen in open water.

Habitat

With the introduction of a wind-turbine foundation, new surfaces for settlement are provided. Depending on factors such as the character of the structure, water dynamics, sedimentation, light availability and species competition, different assemblages may develop (e.g. Paine 1974; Knott *et al.* 2004). Some species such as the blue mussel seemed to have a competitive advantage because their position on the vertical structure allowed for a continuous supply of food, carried by the surrounding waters. This was also reported by Wilhelmsson and Malm (2008) and Maar *et al.* (2009) who found that blue mussels attached to wind-turbine foundations had higher biomass per shell length and higher total biomass, respectively, than did mussels living on the seabed. Zonation patterns along vertical structures penetrating the water column are commonly found (Wildish 1977). This was not observed in the present study, probably because the sample depth range was too narrow. However, zonation patterns were noticed in the general survey, showing a dense algal and barnacle cover down the first 3 m below the water surface followed by mostly blue mussels down to the bottom. The splash-zone part of the foundation may also be of importance because it has been shown to give a non-native species of midge new habitat opportunities (Brodin and Andersson 2009).

In the present study, habitat-forming invertebrates showed preferences for the seabed beneath the foundation. The higher densities of mussels on the seabed could reflect favourable water movement. In addition, heavy mussels may fall down because of difficulties in staying attached to the smooth foundation surface (Qvarfordt *et al.* 2006; Wilhelmsson and Malm 2008). A downward movement of organic debris (both live mussels and faecal matter) from the high-up position on the structure to the seabed is not always beneficial for the benthic ecosystem, and may result in local areas of anoxia when oxygen is used up in the degradation process (Zettler and Pollehne 2006).

Table 2. *P*-values from statistical comparison between the wind-turbine transects (Wilcoxon signed ranks test) and from comparisons between control and turbine transects (Kruskal–Wallis one-way analysis)
Variables are total fish density (*N*), juveniles (*J*) and adults (*A*)

Variable	Wilcoxon signed ranks test			Kruskal–Wallis one-way analysis		
	W–W1	W–W20	W1–W20	W–C	W1–C1	W20–C20
Total fish abundance (*N*)	0.462	0.116	0.345	0.008	0.238	0.669
Total juvenile abundance (*J*)	0.173	0.138	0.598	0.055	0.330	0.104
Total adult abundance (*A*)	0.249	0.138	0.225	0.008	0.747	0.104
Two-spotted goby (*N*)	0.028	0.028	1	0.008	0.469	0.221
Two-spotted goby (*J*)	0.043	0.043	0.180	0.055	0.648	0.221
Two-spotted goby (*A*)	0.028	0.043	0.180	0.008	0.280	0.221
Black goby (*N*)	0.028	0.028	0.115	1	0.453	0.108
Black goby (*J*)	0.042	0.086	0.197	1	0.103	0.328
Black goby (*A*)	0.028	0.028	0.225	1	0.666	0.328

We noticed more blue mussels further away (20 m) from the foundation, and a higher coverage of red algae on the foundations, than what was reported in the study by Wilhelmsson *et al.* (2006*a*), conducted earlier in the same area. The higher coverage of red algae on the foundations could be linked to age (i.e. time of submergence of the structure). Qvarfordt *et al.* (2006) found a correlation between the age of the substratum and the biomass of barnacles and red algae, but not blue mussels, in a study conducted on bridge pillars in the same area as the present study.

The difference in fouling assemblage between the painted steel foundations of the wind turbine and the lighthouse concrete was probably an effect of age and difference in surface heterogeneity (Anderson and Underwood 1994; Andersson *et al.* 2009). Given that the lighthouse foundation was half a century older than the wind turbines, benthic communities would be expected to be in a different stage of succession.

Fish

The present study showed that the wind-turbine foundations sustained more fish compared to surrounding waters. Because the wind turbine provided a vertical structure, new and unique habitat opportunities were created. For example, the flow of water will allow for a continuous stream of food items (plankton) that may be delayed by the small-scale turbulence caused by the structure and protruding parts, favouring planktivorous fish (Neira 2005). Zooplankton is the main food source of two-spotted gobies (Costello *et al.* 1986) and both juveniles and adults were recorded in high densities on and around the wind turbines, especially close to ladders and cable drums.

The fouling assemblage on the vertical surface and seabed could have further habitat effects. For example, mussel shells may be used as nests for spawning two-spotted gobies (cf. Wilhelmsson *et al.* 2006*a*). Indeed, we found gravid two-spotted gobies at three of six foundations and several size classes on all them. The presence of gravid females as well as juveniles, in addition to the overall high abundances, would indicate that the added structure facilitated recruitment of fish (Bohnsack 1989). This was further supported by the fact that the two-spotted goby prefers inshore waters with different habitat characteristics (Wilkins and Myers 1992; Folkestad 2002); these areas were situated 5 km from the wind farm and could supply the wind-farm area with new recruits. In addition, the species has a short life-span which could explain why Wilhelmsson and co-workers (2006*a*) noticed larger shoals with higher densities in the same area 4 years earlier (cf. Magill and Sayer 2002).

Another species that was common in the study area was the black goby. However, it seemed less affected by the presence of wind turbines than were the two-spotted gobies. The quantitative increase of black gobies (both juveniles and adults) close to the foundation, compared with 20 m away, was not statistically significant. The species seemed to show a behavioural response to the wind turbines within the wind-farm area by aggregating close to the foundation, but again, this was not statistically significant. The aggregative behaviour could potentially be facilitated by the high coverage of mussels, which may harbour favourable food items such as amphipods, polychaetes and molluscs (Vesey and Langford 1985; Fjøsne and Gjøsæter 1996). More juvenile black gobies were noticed in the wind-farm area than at the control sites, suggesting that the area around the turbines may function as a nursery area.

Although other fish species were also found in the study area, including sand goby, rock gunnel and viviparous eelpout, none of them seemed affected by the presence of the turbines. In two other studies conducted in Swedish waters in which vertical structures were introduced in an experiment (Wilhelmsson *et al.* 2006*b*; Andersson *et al.* 2009), some species were affected whereas others were less influenced. Species less affected were common in the area having more general preferences in terms of the habitats provided.

In conclusion, our results suggest that the introduction of offshore wind turbines will influence sessile organisms and fish. Blue mussels were the most common sessile organism on the vertical surface of the turbines and on the seabed close to the foundation, whereas 20 m from the structure, red algae dominated. The fouling assemblage on the concrete foundation of the nearby 50-year-old lighthouse was dominated by red algae and hydroids, in contrast to the mussel-dominated assemblage on the 7-year-old wind turbines, indicating that the age and structure of the substratum are also relevant.

It is suggested that offshore wind power will locally affect demersal bottom and semi-pelagic fish species as in the present study, with higher numbers of two-spotted gobies and a behavioural response in black gobies. The effect was probably caused by the introduction of a substrate suitable for shelter, settlement and spawning, as well as enhanced food availability. How fish numbers are affected by the introduction of such resources is determined by either recruitment (production) or redistribution (aggregation). We hypothesise that the increased numbers of two-spotted gobies in the present study were mainly due to enhanced production.

The 'reef effect' might be more obvious in an area with more species or a greater difference between the characteristics of the introduced substrate and surrounding bottom, than was the case in this study. Notably, further research on the ecological effects of offshore wind-power structures in other parts of the world is called for as renewable wind energy becomes more widely adopted.

Acknowledgements

The authors thank the Vattenfall staff at Bergkvara for support during field work, C. Berkström and V. Sandblom for hard work in the field, and M. Asplund for identification of algae. We thank D. Wilhelmsson, C. Berkström, P. Sigray, S. Nylin, H. Kautsky and the anonymous referees for their comments on the manuscript. This study was funded by the EU Sixth Framework program; the DOWNVInD project.

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Manuscript received 22 May 2009, accepted 14 November 2009